

UPM

Neyman – Pearson Lemma : Application

$$H_0 : \mu_0 \quad H_a : \mu_1 ; \mu_1 > \mu_0$$

Assumptions: Normal population with known variance σ

Likelihood function:

$$L = \prod_{i=1}^n \frac{\exp \left[-\frac{1}{2} \left(\frac{x_i - \mu}{\sigma} \right)^2 \right]}{\sqrt{2\pi}\sigma} = \frac{\exp \left[-\frac{1}{2\sigma^2} \sum (x_i - \mu)^2 \right]}{(2\pi\sigma^2)^{n/2}}.$$

Simplification of $\frac{L_1}{L_0} \geq k$ gives an equivalent and simplified definition of critical region

Ack: Hoel, Port, Stone

Neyman – Pearson Lemma : Application

Simplification of $\frac{L_1}{L_0} \geq k$ gives an equivalent and simplified definition of critical region

Since $\mu_1 > \mu_0$:

$$\sum x_i \geq \frac{\sigma^2 \log k}{\mu_1 - \mu_0} + \frac{n(\mu_1 + \mu_0)}{2} \dots\dots\dots (1)$$

$$\forall c \in (0, \infty) \text{ s.t. } \sum x_i \geq c \Rightarrow \exists k \in (0, \infty)$$

Hence any $c \in (0, \infty)$ gives a valid test

Ack: Hoel, Port, Stone

Neyman – Pearson Lemma : Application

$$c = \mu_0 + \Phi^{-1}(1 - \alpha) \frac{\sigma}{\sqrt{n}}$$

Depends on μ_0 and α



Thus for any $\mu_1 > \mu_0$ we get an optimal test according to NP lemma

UPM

Let $\mathcal{C}$ be a class of test for testing $H_0 : \theta \in \Theta_0$ vs $H_a : \theta \in \Theta_0^c$. A test in class $\mathcal{C}$, with power function $\pi(\theta)$, is *uniformly most powerful* (UMP) class $\mathcal{C}$ test if $\pi(\theta) \geq \pi'(\theta)$ for every $\theta \in \Theta_0^c$ and every $\pi'(\theta)$ that is a power function of some test in class $\mathcal{C}$.

But how do we define $\mathcal{C}$?

Class of test

Test function :

$$\phi(x) = P(\text{reject } H_0 | X)$$

Let χ denote the support of X .  Test Statistic

- $R \subseteq \chi$: Rejection region – When $x \in R$, H_0 is rejected
- $R^c \subseteq \chi$: Acceptance region -- When $x \in R^c$, H_0 is not rejected

General
Deterministic
Test Function


$$\phi(x) = \begin{cases} 1 & x \in R \\ 0 & x \in R^c \end{cases}$$

Randomized Test
Function


$$\phi(x) = \begin{cases} 1 & x \in R \setminus R^c \\ \gamma & x \in R \cap R^c \\ 0 & x \in R^c \setminus R \end{cases}$$

Ack: Casella, Berger

Class of test

$$H_0: \theta_0 \quad H_a: \theta_1$$

- Type I error $\alpha = E_{\theta_0} \phi(x)$
- Type II error $\beta = 1 - E_{\theta_1} \phi(x)$

$$\pi(x) = E_{\theta} \phi(x)$$

$$H_0: \theta \in \Theta_0 \text{ vs } H_a: \theta \in \Theta_0^c$$

$$\alpha = \sup_{\theta \in \Theta_0} E_{\theta} \phi(x)$$

$$\mathcal{C} = \{ \text{all } \phi(x) \text{ of level } \alpha \}$$

ex 1

$$\phi(x) = \begin{cases} 1 & x < c \\ 0 & x > c \end{cases}$$

$$\phi(x) = \begin{cases} 1 & x < c \\ \gamma & x = c \\ 0 & x > c \end{cases}$$

ex 2

Monotone likelihood ratio

$f_n(\mathbf{x} \mid \theta)$: joint p. d. f / p. m. f of $X = (X_1, X_2, \dots, X_n)$
where $\mathbf{x} \in \mathbb{R}^n$

$$T = r(\mathbf{x}) \leftarrow \text{Test Statistic}$$

Joint distribution of $\mathbf{X}$ **satisfies MLR in statistic T** if the following holds :

- the ratio $\frac{f_n(\mathbf{x} \mid \theta_1)}{f_n(\mathbf{x} \mid \theta_2)}$ depends on $\mathbf{x}$ through $r(\mathbf{x})$
- for $\theta_1 < \theta_2$ the above ratio is strictly increasing

Monotone likelihood ratio-- Examples

Suppose $X_i \sim \text{ber}(p)$ $i = 1, 2, \dots, n$. Consider $T = \sum_{i=1}^n x_i$.
Here $f_n(\mathbf{t} \mid \theta)$ has MLR property.

Suppose $X_i \sim \mathcal{N}(\mu, \sigma^2)$. Here μ is unknown but σ^2 is known. Consider $T = \frac{1}{n} \sum_{i=1}^n x_i$.
Here $f_n(\mathbf{t} \mid \theta)$ has MLR property.

Suppose $X_i \sim \text{exp}(\theta)$ $i = 1, 2, \dots, n$. Consider $T = \sum_{i=1}^n x_i$.
Is $f_n(\mathbf{t} \mid \theta)$ has MLR property ?